



**ADVANCED COMPUTER NETWORKS
(MTITE101)**

Submitted By: **Aniket Sonker**

Roll no: **2026PIT4227**

Branch: **Information Technology**

Submitted to: **Karan Gupta sir**

Practical 1:

1. Aim: To study network configuration, routing tables, connectivity, packet paths, and IP-to-MAC address mapping using networking commands in the macOS Terminal.

2. Commands Run:

1: `ipconfig /all`

Function: Displays network configuration parameters including local IPv4/IPv6 addresses, MAC address, subnet mask, default gateway, and DNS servers.

Output:

```
Windows PowerShell
PS C:\Users\anike> ipconfig /all

Windows IP Configuration

Host Name . . . . . : AsusAnni123
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-14
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::b92d:2b25:10bd:5da%20(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 688521255
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-6D-12-30-00-0F-C9-36-FE-6A
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter #2
Physical Address. . . . . : 0A-00-27-00-00-09
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::d9b5:4799:8744:2593%9(Preferred)
IPv4 Address. . . . . : 192.168.33.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 772407335
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-6D-12-30-00-0F-C9-36-FE-6A
NetBIOS over Tcpip. . . . . : Enabled

Unknown adapter McAfee VPN:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-BF-D7-18-28
DHCP Enabled. . . . . : Yes
```

2: `route print`

Function: Displays the active IPv4 routing table, including network destinations, netmasks, gateways, active interfaces, and metrics.

Output:

```
Windows PowerShell
PS C:\Users\anike> route print
=====
Interface List
20...0a 00 27 00 00 14 .....VirtualBox Host-Only Ethernet Adapter
9...0a 00 27 00 00 09 .....VirtualBox Host-Only Ethernet Adapter #2
50...00 ff bf d7 18 28 .....TAP-Windows Adapter V9
11...da b3 2f 54 1c 49 .....Microsoft Wi-Fi Direct Virtual Adapter
14...de b3 2f 54 1c 49 .....Microsoft Wi-Fi Direct Virtual Adapter #2
4...d8 b3 2f 54 1c 49 .....Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
5...d8 b3 2f 54 1c 4a .....Bluetooth Device (Personal Area Network)
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          192.168.29.1     192.168.29.57     40
127.0.0.0                  255.0.0.0        On-link          127.0.0.1         331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1         331
127.255.255.255            255.255.255.255  On-link          127.0.0.1         331
192.168.29.0                255.255.255.0    On-link          192.168.29.57     296
192.168.29.57              255.255.255.255  On-link          192.168.29.57     296
192.168.29.255             255.255.255.255  On-link          192.168.29.57     296
192.168.33.0                255.255.255.0    On-link          192.168.33.1      281
192.168.33.1                255.255.255.255  On-link          192.168.33.1      281
192.168.33.255             255.255.255.255  On-link          192.168.33.1      281
192.168.56.0                255.255.255.0    On-link          192.168.56.1      281
192.168.56.1                255.255.255.255  On-link          192.168.56.1      281
192.168.56.255             255.255.255.255  On-link          192.168.56.1      281
224.0.0.0                  240.0.0.0        On-link          127.0.0.1         331
224.0.0.0                  240.0.0.0        On-link          192.168.56.1      281
224.0.0.0                  240.0.0.0        On-link          192.168.33.1      281
224.0.0.0                  240.0.0.0        On-link          192.168.29.57     296
255.255.255.255            255.255.255.255  On-link          127.0.0.1         331
255.255.255.255            255.255.255.255  On-link          192.168.56.1      281
255.255.255.255            255.255.255.255  On-link          192.168.33.1      281
255.255.255.255            255.255.255.255  On-link          192.168.29.57     296
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
4         56 :::/0                fe80::4e22:f3ff:fee4:b4e8
```

3: ping -n 4 8.8.8.8

Function: Sends ICMP Echo Request packets (4 requests specified by -n 4) to test connectivity, round-trip times, and packet loss.

Output:

```
Windows PowerShell
PS C:\Users\anike> ping -n 4 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=12ms TTL=112
Reply from 8.8.8.8: bytes=32 time=15ms TTL=112
Reply from 8.8.8.8: bytes=32 time=9ms TTL=112
Reply from 8.8.8.8: bytes=32 time=16ms TTL=112

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 16ms, Average = 13ms
PS C:\Users\anike> |
```

4: `tracert 8.8.8.8`

Function: Traces the path and measures latency across each intermediate router hop to reach the target IP.

Output:

```
Windows PowerShell
PS C:\Users\anike> tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1    3 ms    10 ms    2 ms  reliance.reliance [192.168.29.1]
  2    7 ms    12 ms    5 ms  10.11.240.1
  3    8 ms     8 ms    8 ms  172.31.5.141
  4   17 ms     9 ms   63 ms  192.168.128.138
  5    6 ms    10 ms    7 ms  172.27.248.55
  6    7 ms    10 ms    9 ms  172.27.248.34
  7   11 ms    10 ms    8 ms  192.168.44.24
  8    *      *      *    Request timed out.
  9    *      *      *    Request timed out.
 10    9 ms    21 ms   69 ms  72.14.195.34
 11   37 ms    16 ms    9 ms  108.170.237.85
 12   10 ms     9 ms    8 ms  142.251.52.213
 13    9 ms    23 ms    8 ms  dns.google [8.8.8.8]

Trace complete.
PS C:\Users\anike>
```

5: `arp -a`

Function: Displays the Address Resolution Protocol (ARP) cache, showing current IPv4-to-MAC address mappings on your local network.

Output:

```
Windows PowerShell
PS C:\Users\anike> arp -a

Interface: 192.168.29.57 --- 0x4
Internet Address      Physical Address      Type
-----
192.168.29.1          4c-22-f3-e4-b4-e8    dynamic
192.168.29.99         9c-30-5b-b2-ed-25    dynamic
192.168.29.135        18-eb-d4-cf-b7-19    dynamic
192.168.29.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
232.24.3.241         01-00-5e-18-03-f1    static
239.255.255.250      01-00-5e-7f-ff-fa    static
255.255.255.255      ff-ff-ff-ff-ff-ff    static

Interface: 192.168.33.1 --- 0x9
Internet Address      Physical Address      Type
-----
192.168.33.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
232.24.3.241         01-00-5e-18-03-f1    static
239.255.255.250      01-00-5e-7f-ff-fa    static

Interface: 192.168.56.1 --- 0x14
Internet Address      Physical Address      Type
-----
192.168.56.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
232.24.3.241         01-00-5e-18-03-f1    static
239.255.255.250      01-00-5e-7f-ff-fa    static
PS C:\Users\anike>
```

3. Result: The networking commands were successfully executed, and the network interface configuration, IPv4 routing table, connectivity, network path, and ARP mappings were successfully observed and analyzed.